

PAPOLLO HEALTHCARE DASHBOARD

Healthcare Operations, Revenue & Patient Analytics Report

Project Detail	Information
Project	Papollo Healthcare Dashboard
Domain	Healthcare Operations & Patient Analytics
Tool	Tableau
Analysis Areas	Patients, diagnosis, billing, insurance, tests, doctors and length of stay
Dashboard Focus	Operational efficiency, revenue leakage and patient utilization
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1. Executive Summary

The Papollo Healthcare Dashboard is a healthcare analytics project designed to examine patient activity, diagnosis patterns, billing and health-insurance amounts, diagnostic-test preferences, doctors associated with patient records, and average length of stay. The dashboard combines operational and financial views so that hospital management can identify utilization patterns and potential revenue leakage.

The supplied visuals show that Viral Infection has the largest displayed billing amount and insurance amount, followed by Flu and Malaria. Blood Test is the most frequently preferred test in the displayed ranking. Average length of stay is highest for Malaria, followed by Viral Infection and Fracture.

2. Business Problem

Healthcare organizations need visibility into patient volumes, treatment patterns, resource utilization and financial performance. The dashboard addresses this by bringing operational and financial indicators into one analytical view.

- Track patient activity by year of admission.
- Compare billing and health-insurance amounts across diagnoses.
- Identify diagnoses associated with longer hospital stays.
- Understand the most frequently preferred diagnostic tests.
- Compare doctor-level patient counts as an operational indicator.
- Support investigation of potential revenue leakage.

3. Dashboard Overview

Dashboard Component	Purpose	Key Observation from Supplied Visual
Patient By Years	Compare average length of stay across admission years and bed occupancy categories.	2022 has the largest displayed stacked total; 2023 and 2024 are substantially lower.
Revenue Leakage	Compare billing and health-insurance amounts by diagnosis.	Viral Infection has the highest displayed amounts; Fracture has the lowest.
Average Length of Stay	Rank diagnoses by average stay.	Malaria is highest at about 8.4 days; Viral Infection is about 8.3 days.
Operational Efficiency	Compare patient counts associated with doctors.	Displayed doctor counts are all close to 1,000 patients.
Most Test Dr. Prefer	Rank diagnostic tests by patient count.	Blood Test ranks first, followed by MRI and CT Scan.

4. Visual Analysis

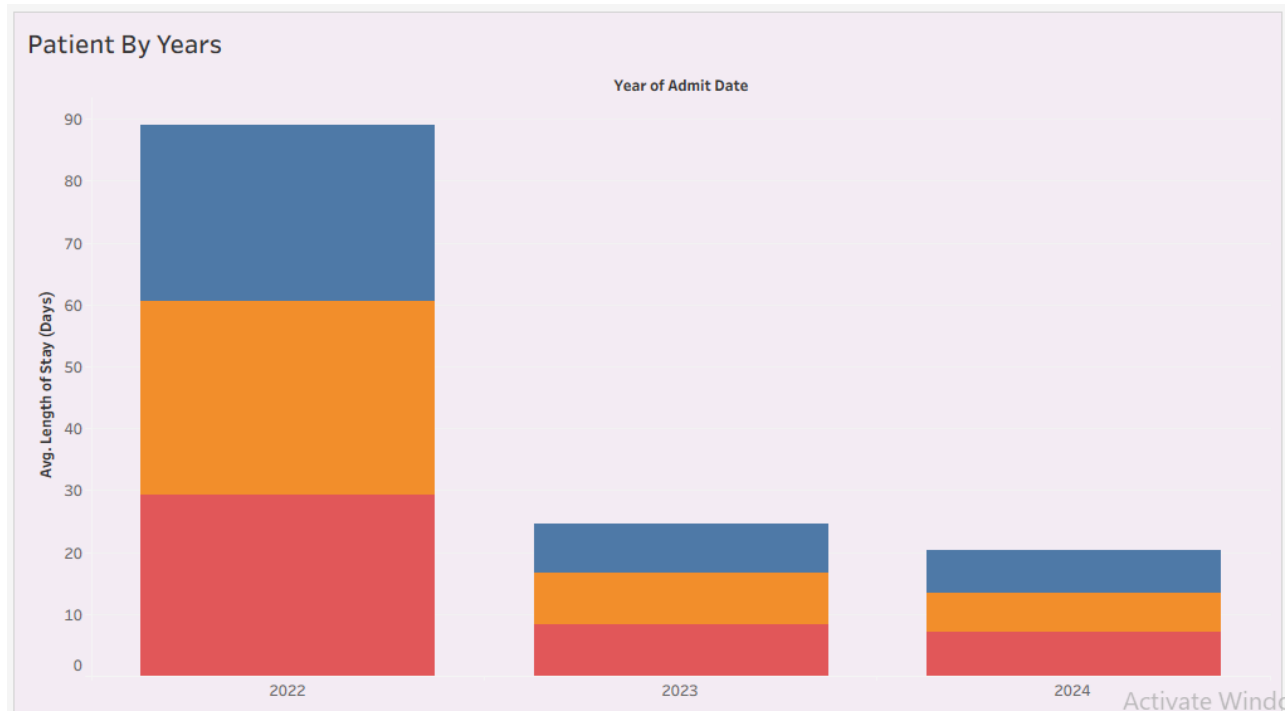


Figure 1. Patient by Years

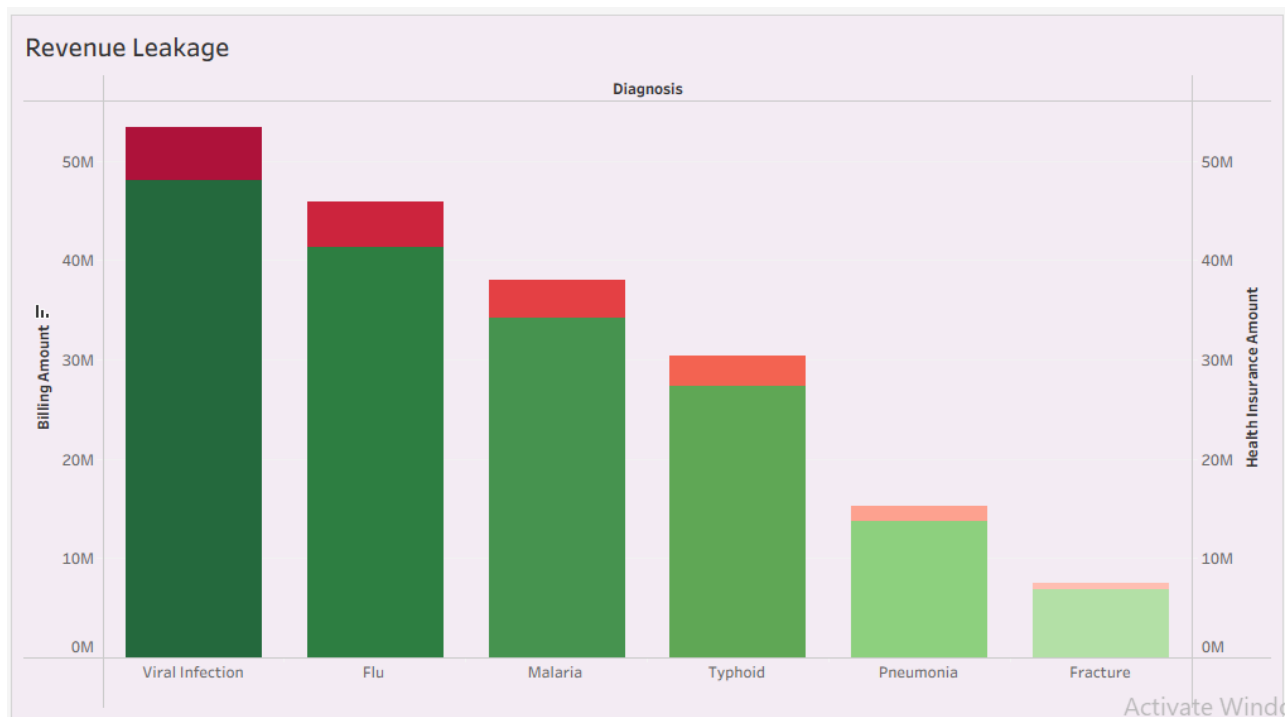


Figure 2. Revenue Leakage

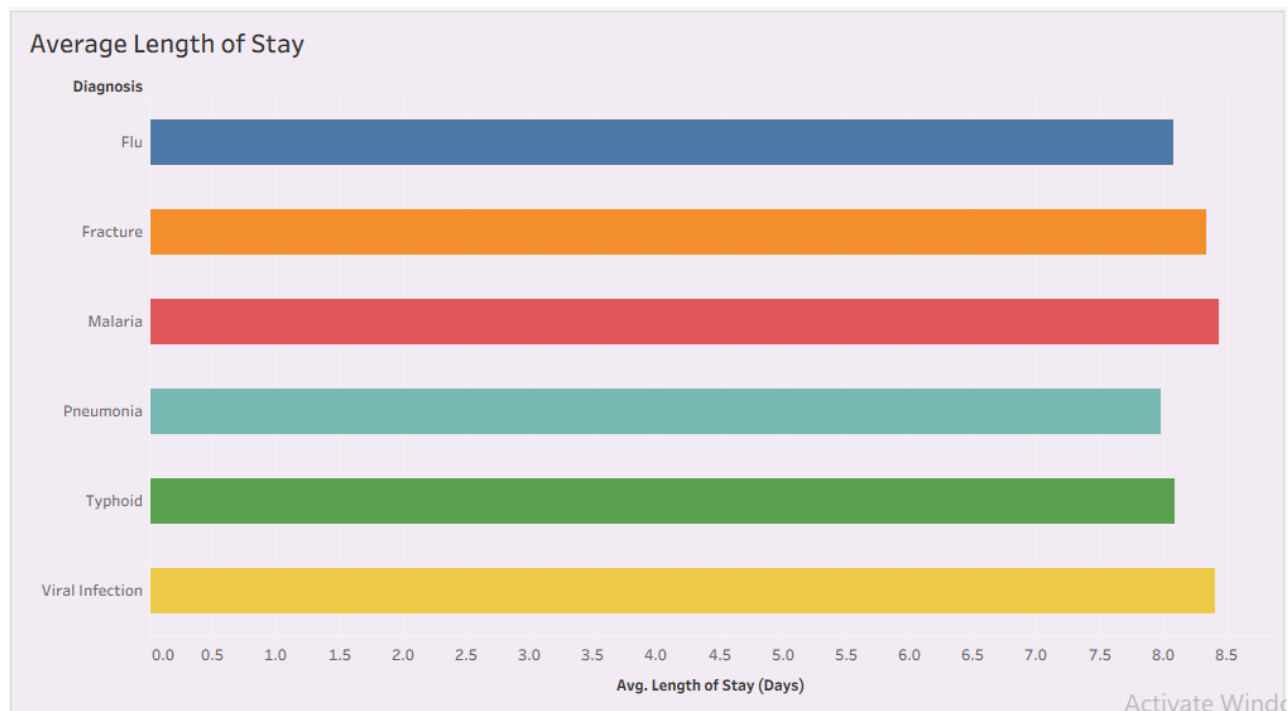


Figure 3. Average Length of Stay

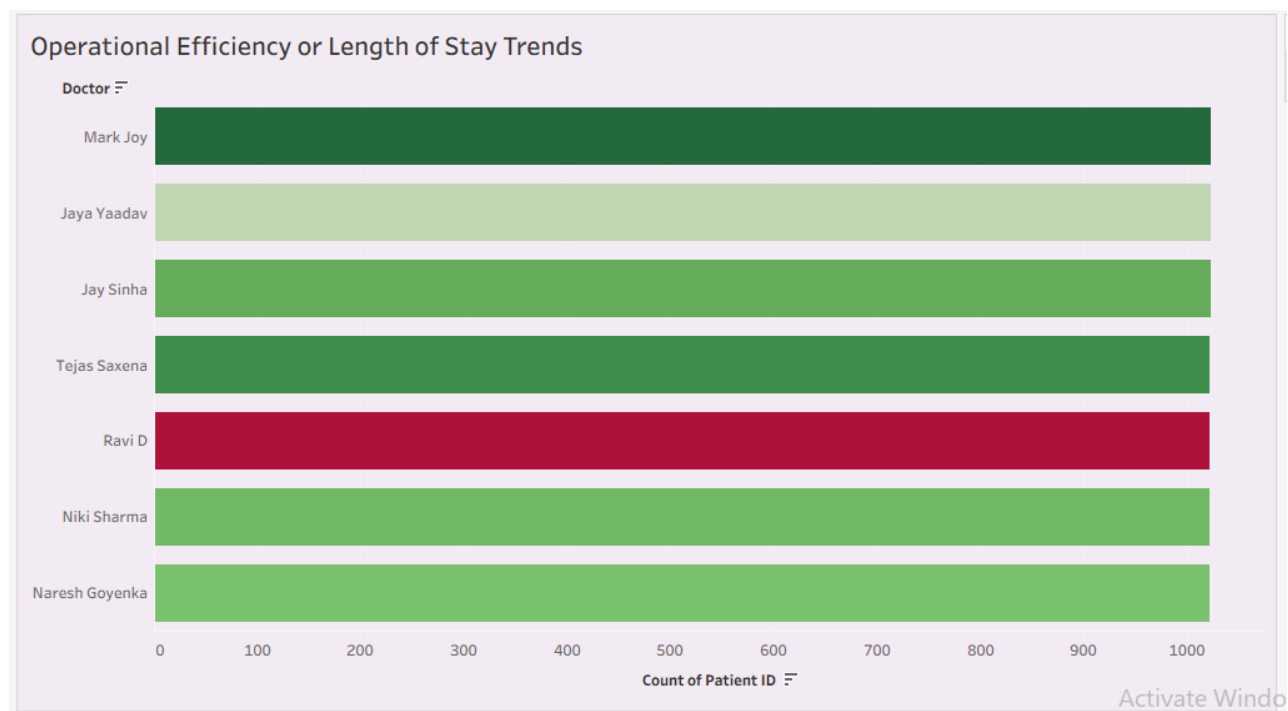


Figure 4. Operational Efficiency / Length of Stay Trends

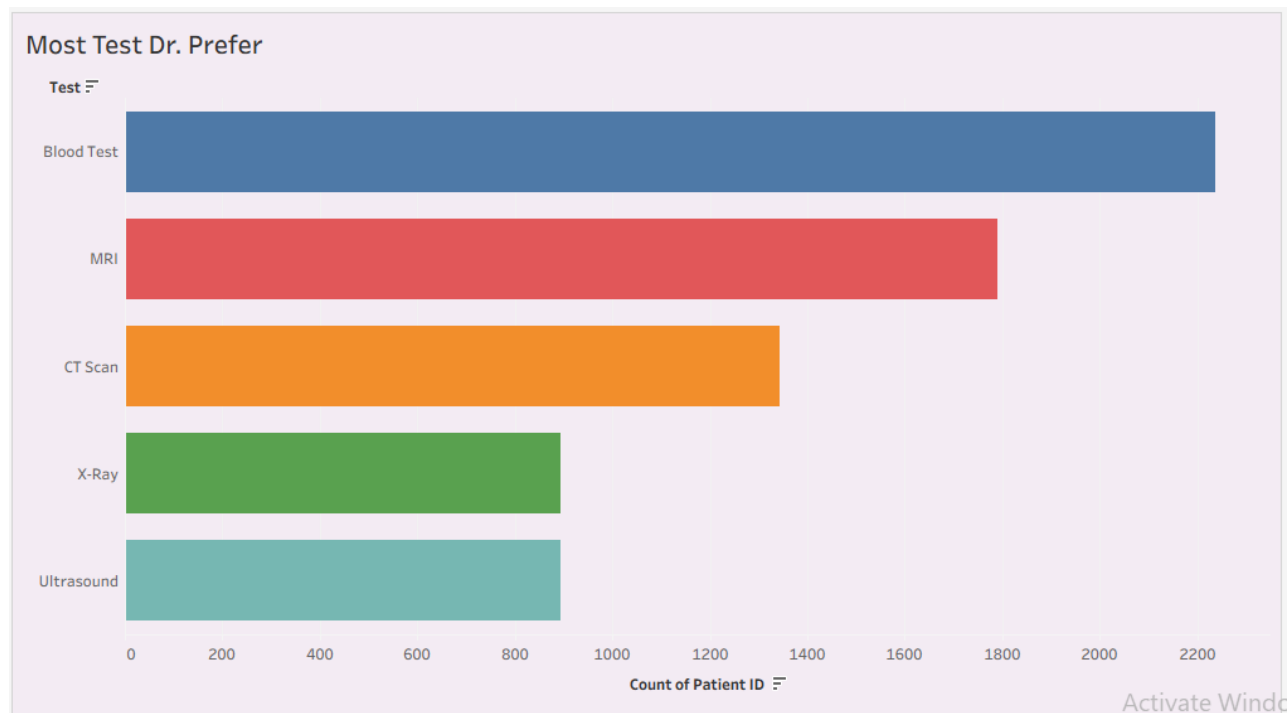


Figure 5. Most Test Dr. Prefer

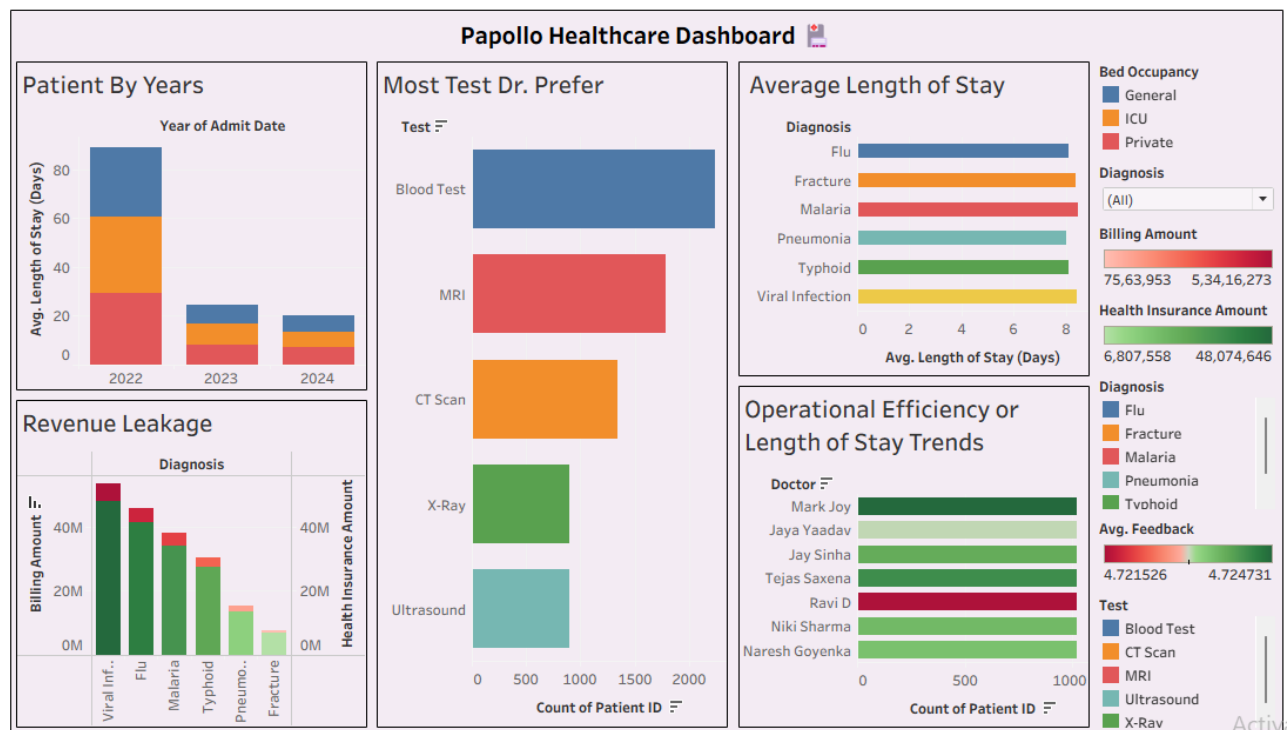


Figure 6. Combined Papallo Healthcare Dashboard

5. Patient Admission & Length of Stay Analysis

The Patient By Years chart shows a clear reduction in the displayed average-length-of-stay totals from 2022 to 2024. The 2022 stacked bar reaches roughly 89 days, while 2023 is around 25 days and 2024 is around 20 days. The stacked colors represent bed-occupancy categories shown in the dashboard.

This should be interpreted carefully: the chart is labeled 'Avg. Length of Stay (Days)' and is stacked by bed occupancy. Therefore, the visual should not be described simply as the number of patients admitted per year.

6. Revenue Leakage Analysis

Diagnosis	Relative Billing Amount	Relative Insurance Amount	Observation
Viral Infection	Highest	Highest	Largest financial volume in the visual.
Flu	2nd highest	2nd highest	Large financial contribution.
Malaria	3rd highest	3rd highest	Material financial volume.
Typhoid	Mid-range	Mid-range	Lower than Malaria, above Pneumonia/Fracture.
Pneumonia	Lower	Lower	Smaller financial volume.
Fracture	Lowest	Lowest	Smallest displayed financial volume.

The Revenue Leakage chart compares Billing Amount and Health Insurance Amount by diagnosis. The large differences between diagnosis categories identify where financial-volume analysis should be concentrated. However, the screenshot alone does not provide a leakage percentage or prove that the gap between billing and insurance represents recoverable leakage.

Portfolio wording recommendation: describe this as 'revenue leakage analysis' or 'potential revenue leakage' unless the underlying model explicitly defines a leakage measure.

7. Average Length of Stay by Diagnosis

Diagnosis	Approx. Average Stay	Relative Position
Malaria	8.4 days	Highest
Viral Infection	8.3 days	Very high
Fracture	8.3 days	Very high
Typhoid	8.1 days	High
Flu	8.1 days	High
Pneumonia	8.0 days	Lowest of the displayed group

The differences are relatively small—roughly 8.0 to 8.4 days—so the dashboard suggests that average stay is fairly consistent across the six diagnoses, with Malaria at the top of the displayed ranking.

8. Diagnostic Test Preference

Blood Test is the most frequently preferred test in the displayed ranking, with approximately 2,250 patient records. MRI follows at approximately 1,750, CT Scan at approximately 1,300, and X-Ray and Ultrasound at approximately 900 each.

- Blood Test is the dominant diagnostic-test category.
- MRI is the second-highest category.
- CT Scan has a substantial but lower patient count.

- X-Ray and Ultrasound have similar displayed counts.
- These values represent counts of Patient ID in the visualization; they should not automatically be interpreted as the number of unique tests performed unless the underlying measure is DISTINCTCOUNT.

9. Doctor / Operational Efficiency Analysis

The Operational Efficiency or Length of Stay Trends chart shows patient counts by doctor. The displayed values are all close to 1,000, with only modest differences among the listed doctors.

This chart is useful as a workload indicator, but patient count alone is not enough to measure clinical efficiency. A stronger operational KPI would combine patient volume with average length of stay, case mix, readmissions, outcomes and resource utilization.

10. Combined Dashboard Interpretation

The combined dashboard provides a compact management view of the major analytical areas: patient-year patterns, revenue leakage, test preference, average length of stay and doctor-level workload. Filters shown on the dashboard include Diagnosis, Billing Amount, Health Insurance Amount, Bed Occupancy, and other dimensions.

This structure is appropriate for a healthcare analytics portfolio because it demonstrates both descriptive analysis and an attempt at diagnostic business analysis.

11. Key Business Findings

- Viral Infection has the largest displayed billing and health-insurance amounts.
- Fracture has the smallest displayed financial amounts among the six diagnoses.
- Malaria has the highest displayed average length of stay at approximately 8.4 days.
- The displayed average stays for all diagnoses are tightly clustered around 8 days.
- Blood Test is the most frequently preferred diagnostic test.
- MRI is the second most frequently preferred test in the displayed ranking.
- Doctor-level patient counts are broadly similar, suggesting a relatively balanced displayed workload.
- The 2022 Patient By Years visual is substantially larger than the 2023 and 2024 visuals.

12. Business Recommendations

1. Investigate high-volume diagnoses

Review Viral Infection, Flu and Malaria for billing, insurance reimbursement and treatment-cost patterns.

2. Monitor long-stay diagnoses

Malaria and other higher-stay diagnoses should be investigated for care pathways, resource utilization and discharge planning.

3. Analyze insurance gaps

Compare billed amounts, insured amounts, approved amounts and collected amounts if those fields are available in the source data.

4. Optimize diagnostic resource planning

Use the test-preference ranking to plan laboratory and imaging capacity, especially for Blood Tests and MRI.

5. Use workload KPIs carefully

Combine patient volume with case complexity, length of stay and outcomes before making doctor-efficiency conclusions.

6. Build a true leakage KPI

If the source data supports it, calculate leakage as an explicit monetary or percentage measure rather than inferring it from two separate totals.

13. Data Quality & Validation Notes

- This report is based on the supplied dashboard screenshots; the underlying Tableau workbook and row-level dataset were not independently audited in this response.
- The 'Revenue Leakage' visual does not, by itself, prove that the difference between billing and insurance is leakage.
- The 'Operational Efficiency' chart uses patient count, which is a workload measure rather than a complete efficiency metric.
- The Patient By Years chart uses average length of stay and should not be described as patient volume without checking the underlying measure.
- Exact numerical values should be taken from the underlying Tableau data when publishing a final report, because several screenshot labels are rounded or visually estimated.

14. Technical Skills Demonstrated

- Tableau dashboard development
- Healthcare data visualization
- KPI and metric design
- Patient and diagnosis analysis
- Revenue and insurance analysis
- Length-of-stay analysis
- Diagnostic-test utilization analysis
- Operational workload analysis
- Interactive filters
- Business storytelling and recommendations

15. Conclusion

The Papollo Healthcare Dashboard is a strong portfolio project because it connects patient operations with financial and utilization analysis. The dashboard highlights high-volume diagnoses, length-of-stay patterns, diagnostic-test demand and doctor-level workload.

The most important improvement before presenting this project professionally is to clearly define the Revenue Leakage metric and validate the underlying measures. With that clarification, the project can demonstrate not only dashboarding skills but also careful healthcare business reasoning.